

Computational Identification of Unknown Organic Contaminants and Molecules in Remote Environments via Molecular Networking

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1 Goal

This project aims to characterize surface water samples from remote high-altitude environments by identifying both known and unknown organic contaminants using high-resolution mass spectrometry (HRMS). While database-driven approaches with lists of known contaminants enable identification of some contaminants, I hypothesize that a significant fraction of uncharacterized molecules can be identified through non-targeted analysis, particularly molecular networking. Together, these approaches will reveal the identity and patterns of contaminants in these remote regions.

2 Specific Aims

2.1 Aim 1: Preparation and Measurement of Surface Water Samples

- **What (Aim):** Measure organic contaminants in surface water samples collected from remote high-altitude regions, including Mt. Everest, Mt. Kilimanjaro, the Tibetan Plateau, and Qinghai Lake.
- **How (Approach):** Samples will be analyzed using iterative data-dependent acquisition (DDA), where each sample is run multiple times to increase feature detection and improve coverage of low-abundance compounds.

- **Impact:** This aim will generate high-quality data with many mass spectral features, which will be subjected to identification and analysis of contaminants.

2.2 Aim 2: Identify contaminants using suspect screening with the help of Agilent environment

- **What (Aim):** Identify known environmental contaminants, pharmaceuticals, and other anthropogenic compounds using suspect screening (Agilent chemical database).
- **How (Approach):** HRMS data will be compared against curated chemical databases (e.g., Agilent database) using software such as MS-DIAL, leveraging exact mass and MS/MS spectral matching for compound annotation.
- **Impact:** This aim will establish a baseline of known contaminants present without additional identification efforts, allowing for comparison with non-targeted approaches.

2.3 Aim 3: Characterize unknown contaminants using molecular networking

- **What (Aim):** Identify and characterize unknown organic molecules not captured by database screening using molecular networking approaches.
- **How (Approach):** MS/MS spectral similarity will be used to construct molecular networks, allowing unknown compounds to be grouped with structurally related known molecules. Mass differences and fragmentation patterns will be analyzed to propose potential structures and transformation products.
- **Impact:** This aim will fill critical gaps in contaminant identification by enabling discovery of novel compounds and environmental transformation products, significantly expanding understanding of chemical pollution in remote ecosystems.

References

- Fan, W. Suspect Screening and Semi-Quantification of Organic Contaminants in Snow and Water Samples on the Qinghai–Tibet Plateau and High Mountain Regions. Undergraduate honors thesis, The College of Wooster, Wooster, OH, 2025; Unpublished thesis.
- Kind, T.; Tsugawa, H.; Cajka, T.; Ma, Y.; Lai, Z.; Mehta, S. S.; Wohlgemuth, G.; Barupal, D. K.; Showalter, M. R.; Arita, M.; Fiehn, O. *Mass Spectrometry Reviews* **2018**, *37*, 513–532.
- Schymanski, E. L.; Jeon, J.; Gulde, R.; Fenner, K.; Ruff, M.; Singer, H. P.; Hollender, J. *Environmental Science & Technology* **2014**, *48*, 2097–2098.
- Ye, W.; Li, J.; Cai, X. *Journal of Computational Chemistry* **2025**, *46*, e70011.
- Manz, K. E.; Feerick, A.; Braun, J. M.; Feng, Y. L.; Hall, A.; Koelmel, J.; Manzano, C.; Newton, S. R.; Pennell, K. D.; Place, B. J.; Godri Pollitt, K. J.; Prasse, C.; Young, J. A. *Journal of Exposure Science & Environmental Epidemiology* **2023**, *33*, 524–536.
- Nino-Suastegui, S.; Painter, E.; Sprankle, J. W.; Morrison, J. J.; Faust, J. A.; Gray, R. *Environmental Research* **2025**, *266*, 120494.
- Chao, A.; Minucci, J. M.; Ferland, T. M.; Carr, E. T.; Janesch, G.; Rizwan, S.; Whitehead, H. D.; Cathey, T.; Pu, S.; Brunelle, L. D.; Batt, A. L.; Sobus, J. R.; Williams, A. J. *Analytical Chemistry* **2025**, *97*, 15904–15912.
- Schmid, R.; Petras, D.; Nothias, L.-F.; Wang, M.; Aron, A. T.; Jagels, A.; Tsugawa, H.; others *Nature Communications* **2021**, *12*, 3832.
- Ernst, M.; Kang, K. B.; Caraballo-Rodríguez, A. M.; Nothias, L.-F.; Wandy, J.; Chen, C.; Wang, M.; others *Metabolites* **2019**, *9*, 144.
- Olivon, F.; Grelier, G.; Roussi, F.; Litaudon, M.; Touboul, D. *Analytical Chemistry* **2017**, *89*, 7836–7840.
- Zhu, W.; Zhang, Y.; Zhao, D.; Xu, J.; Wang, L. *Journal of Chemical Information and Modeling* **2023**, *63*, 43–55.
- Herbert, B. M. J.; Villa, S.; Halsall, C. J. *Ecotoxicology and Environmental Safety* **2006**, *63*, 3–16.
- Daly, G. L.; Wania, F. *Environmental Science & Technology* **2005**, *39*, 385–398.
- Blais, J. M., Rosen, M. R., Smol, J. P., Eds. *Environmental Contaminants: Using Natural Archives to Track Sources and Long-Term Trends of Pollution*; Springer: Dordrecht, 2015.

- Xu, Y.; Zhang, G.; Li, J.; Chakraborty, P.; Li, H.; Liu, X. *Journal of Environmental Monitoring* **2011**, *13*, 3119–3127.
- Kiss, A.; Fries, E. *Environmental Science and Pollution Research* **2009**, *16*, 702–710.
- Gray, R.; Painter, E.; Sprankle, J. W.; Crawford, A. D.; Morrison, J. J.; Frazier, M.; Faust, J. A. *Atmospheric Environment* **2022**, *291*, 119389.
- Wang, M.; Carver, J. J.; Phelan, V. V.; Sanchez, L. M.; Garg, N.; others *Nature Biotechnology* **2016**, *34*, 828–837.
- Fisher, P. K. N. S. e. a., C.M. *Anal Bioanal Chem* **2022**, *414*, 6455–6471.
- Catalan, J., Ed. *Tracking Long-Range Atmospheric Transport of Trace Metals, Polycyclic Aromatic Hydrocarbons, and Organohalogen Compounds Using Lake Sediments of Mountain Regions*; Springer: Dordrecht, 2015.
- Asnicar, F.; Thomas, A. M.; Passerini, A.; Waldron, L.; Segata, N. *Nature Reviews Microbiology* **2024**, *22*, 191–205.
- Ma, Z.; Meng, S.; Gao, X.; Wan, H.; Fang, L.; Zhu, L.; Li, Z.; Gao, Y.; Qiu, L.; Rong, B.; Song, C. *Journal of Hazardous Materials* **2025**, *496*, 139493.
- Hupatz, H.; Rahu, I.; Wang, W.-C.; others *Analytical and Bioanalytical Chemistry* **2025**, *417*, 473–493.
- Schulze, B.; Jeon, Y.; Kaserzon, S.; Heffernan, A. L.; Dewapriya, P.; O’Brien, J.; Gomez Ramos, M. J.; Ghorbani Gorji, S.; Mueller, J. F.; Thomas, K. V.; Samanipour, S. *TrAC Trends in Analytical Chemistry* **2020**, *133*, 116063.
- Koelmel, J.; Kroeger, N.; Gill, E.; Ulmer Holland, C.; Bowden, J.; Patterson, R.; Yost, R.; Garrett, T. *Journal of The American Society for Mass Spectrometry* **2017**, *28*, 1–10.
- Tsugawa, H.; Cajka, T.; Kind, T.; Ma, Y.; Higgins, B.; Ikeda, K.; Kanazawa, M.; VanderGheynst, J.; Arita, M. *Nature methods* **2015**, *12*.
- Zahn, D.; Arp, H. P. H.; Fenner, K.; Georgi, A.; Hafner, J.; Hale, S. E.; Hollender, J.; Letzel, T.; Schymanski, E. L.; Sigmund, G.; Reemtsma, T. *Environmental Science & Technology* **2024**, *58*, 7710–7718.
- Bittremieux, W.; Schmid, R.; Huber, F.; van der Hooft, J. J. J.; Wang, M.; Dorrestein, P. C. *Journal of the American Society for Mass Spectrometry* **2022**, *33*, 1733–1744.
- Dalla Valle, N.; Garcia-Aloy, M.; Robatscher, P.; Franceschi, P.; Oberhuber, M. *Analytical Chemistry* **2025**, *97*, 15873–15882, PMID: 40673560; PMCID: PMC12311886.